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Roll No.: _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

Amrita Vishwa Vidyapeetham

Int MSc/BSc Degree Examination – June 2024

Second Semester

Integrated MSc Physics with Minor in Scientific Computing/Data Science

Electricity and Magnetism

22PHY114

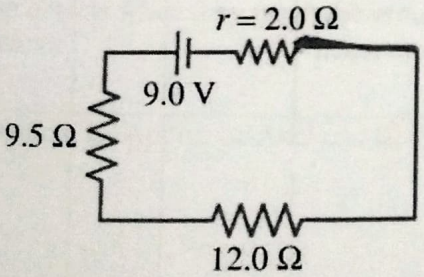
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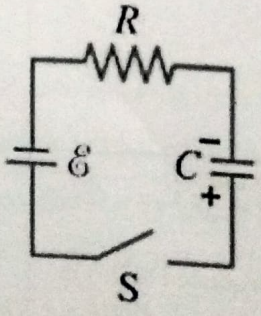
Time :3 hours

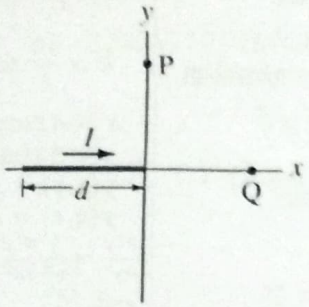
Maximum Marks:100

Instruction

1. Answer all the questions.
2. Answer with appropriate significant figures and units.

Qn No.		Marks	CO	Blooms Taxonomy level
1	A wire of length 2.0 m carries 8.2 A. A magnetic field, **B** is applied. If the wire lies along the +x-axis, a magnetic force **F** = −2.5 **j** N acts on the wire. If the wire lies along the +y-axis, a magnetic force **F** = (2.5 **i** – 5 **k**) N acts on the wire. Find **B**. You can leave the answer in terms of the unit vectors.	10	CO4	L3`
2	For the circuit shown in the figure below, a) Find the current in the circuit. b) Using the current determined in part (a), show that sum of all the voltage changes around the circuit is zero. 	10	CO3	L3
3	Stream of protons pass a given point in space at a rate of 2.5×10^9 protons/s. a) What is the current due to the protons? b) What is the magnitude of the magnetic field produced at 2.0 m from the beam of protons?	10	CO4	L3

4	In an LRC circuit connected to a 10.0 kHz, 725 V (rms) source, the inductance is 32.0 mH, resistance is 8.70 kΩ while the capacitance is 6.25 nF. a) What is the inductive reactance, capacitive reactance, and total impedance in the circuit? b) What is the phase angle? c) What is the rms current?	10	CO5	L3
5	There are two charges, −8.0 μC and 5.8 μC which are 8 cm apart. Find the magnitude and direction of the electric field at the mid-point between the two charges. You can assume that the charges lie on the x-axis.	10	CO1	L3
6	The area of a flexible circular loop at $t = 0$ is 0.285 m^2. The Area decreases at a rate such that $dA/dt = -3.50 \times 10^{-2}$ m^2/s. There is a uniform magnetic field 0.280 T applied perpendicular to the plane of the loop. a) Determine the induced emf at $t = 0$. b) What is the area at $t = 2.00$ s? c) What is the induced emf at $t = 2.00$ s?	10	CO5	L3
7	A 425×10^{-12} F capacitor is charged by a 135 V source before it is quickly connected to a 0.175 H inductor. Given that $q = q_{max}\cos(\omega t + \theta)$ where the symbols have their usual meanings with SI units. a) Determine the frequency, f, in the circuit. b) Find the maximum current in the LC circuit. c) Find the maximum energy stored by the inductor.	10	CO5	L3
8	In the circuit below, $R = 5.0 \times 10^3$ Ω, while the emf of the battery (ε) is 12.0 V. The time constant (τ) is 25.0×10^{-6} s. a) Find the total capacitance of the circuit. b) Find the time it takes for the voltage across the resistor to reach 4.0 V. Given that the voltage across the capacitor, $V_c = \varepsilon(1 - e^{(-t/\tau)})$. 	10	CO3	L3

9	A straight wire of length d carries a current I and lies on the negative x-axis as shown in the figure.  a) What is the magnetic field $\boldsymbol{B}$ at point Q that is along the axis of the wire? Why? b) Show that the magnetic field at a point P is $$\frac{\mu_0 I}{4\pi y}\frac{d}{\left(y^2+d^2\right)^{1/2}}\hat{\mathbf{k}}$$ Hint: You can choose a length $d\mathbf{l} = dx\,\mathbf{i}$ and $\boldsymbol{r} = -x\mathbf{i} + y\mathbf{j}$ and note that $$\int \frac{dx}{\left(x^2+y^2\right)^{3/2}} = \frac{x}{y^2\left(x^2+y^2\right)^{1/2}}$$	10	CO4	L3
10	Two charged objects $q_1 = 450 \times 10^{-6}$ C and $q_2 = -450 \times 10^{-6}$ C are dropped from a height of 2.00m. The mass of both the objects is 0.340 kg. There is an uniform electric field of 150 V/m near the surface of the earth whose direction is downwards. Using the law of conservation of energy or otherwise, determine the difference of the speeds of the objects when they reach the ground. You can neglect air resistance.	10	CO2	L3

L1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create